



Digital Image Processing

INSTRUCTOR: YAN-TSUNG PENG

DEPT. OF COMPUTER SCIENCE, NCCU

CLASS 1

Syllabus

- ❑ Time: Wed (13:10 – 16:00)
- ❑ Location: 大勇210207
- ❑ Textbook: Digital Image Processing by Gonzalez and Woods. 4th edition, Pearson
- ❑ TAs:
 - ❑ 陳子昊 (113761501@g.nccu.edu.tw)、廖家樂 (110703048@g.nccu.edu.tw)、李胤寬 (109701002@g.nccu.edu.tw)
 - ❑ Office Hours: 12:00~13:00 Wed.
- ❑ References:
 1. Digital Image Processing Using Matlab, 2nd edition, 2009
 2. Introduction to Digital Image Processing, William K. Pratt, 1st edition, CRC Press, 2013
 3. Computer Vision: Algorithms and Applications, Springer-Verlag, 2011

NCCU Moodle Platform

□ Website: <https://moodle-course02.nccu.edu.tw/course/view.php?id=5885>

□ 1132_753833001_影像處理

- **Materials:** Course slides
- **Grading:** Assessment criteria and grading details
- **Homework:** Submit reports via the course portal

Overview

- ❑ With the rapid advancement of **smartphone camera technology**, high-quality photography is now accessible to everyone. In 2018, an estimated **350 million photos** were uploaded to Facebook daily, a number that has grown with platforms like Instagram and TikTok.
- ❑ Every image and video undergoes **Image Signal Processing (ISP)**, integrating **noise reduction, sharpening, gamma correction, and compression**. Recent developments include **AI-driven enhancement, super-resolution, real-time stabilization, and deep learning-based restoration**, improving low-light performance, color accuracy, and computational editing.
- ❑ Emerging technologies like **NeRF, GANs, and diffusion models** are reshaping **computational imaging, medical imaging, autonomous driving, and AR**, driving the future of image and video processing.

Goals

- ❑ Students will explore a range of image and video processing technologies, gaining a deeper understanding of modern imaging techniques. Additionally, they will engage in hands-on applications, applying image processing methods to solve real-world problems.

Grading Policy

- ❑ Homework (30%) – Three assignments (programming + report).
- ❑ Midterm (30%)
- ❑ Final Presentation (40%)
 - Select a recently published paper from IEEE journals or top-tier conferences and study it.
 - Present your chosen paper at the end of the semester.
 - Tips for a higher score:
 - Submit a full report.
 - Provide experimental results by running the code (either obtained from the authors or implemented by yourself) – code submission required.
 - Propose a "better" method – code submission required.
- ❑ Class Participation (10%) – Quiz
 - Awarded to students present in class and taking the quiz. **No exceptions.**

Grading Policy

- ❑ Homework must be done independently.
 - You may use C/C++, Python, Matlab, etc. If plagiarism is detected, both the copier and the original author will share the total points evenly.
 - Homework submissions must include both code and a report.
 - **Late Submission Penalty:**
 - Up to 3 days late: 70% of the points
 - Up to 6 days late: 50% of the points
 - 7 days or more: No points awarded

Final Project Guidelines

- ❑ Team Formation: You may work in teams of up to 3 members for the final project (not homework)
- ❑ Grading Criteria: Grades will be based on effort and contribution to the project
- ❑ Final Report: Must be submitted by the end of Week 18 (no late submissions)
- ❑ Topic Selection: Projects can be related to, but are not limited to, topics covered in class:
 - ❑ Image Acquisition, Image Enhancement, Image Restoration, Image Compression, Morphological Processing, Segmentation, Classification, Recognition, etc.

Suggested Journals & Conferences

❑ Journals:

- IEEE Transactions on Image Processing
- IEEE Transactions on Pattern Analysis and Machine Intelligence
- IEEE Transactions on Multimedia
- International Journal of Computer Vision (IJCV)

❑ Conferences:

- CVPR, ECCV, ICCV, ACCV, WACV, BMVC
- AAAI, IJCAI, NeurIPS
- IJCV, CVIU, ICML

You are free to choose any topics and papers related to the course. If your selected papers are not from the list above, feel free to discuss them with me.

Report Format

☐ **Full Report (Full Credit):**

Must include the following sections:

- a) Abstract
- b) Introduction
- c) Method
- d) Experiments
- e) Conclusion
- f) References

☐ **Alternative Format (Up to 70% Credit):**

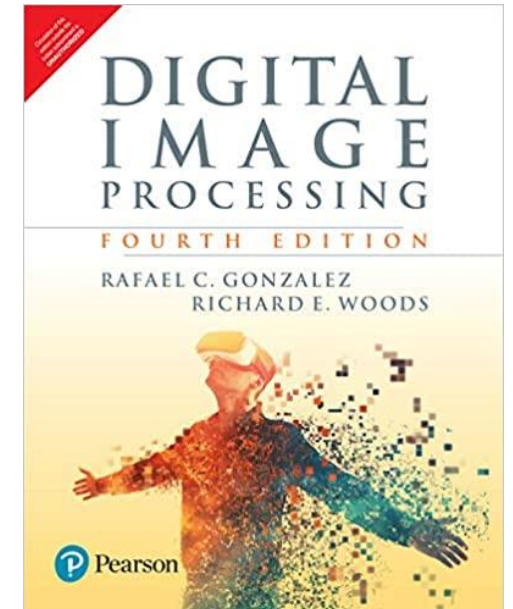
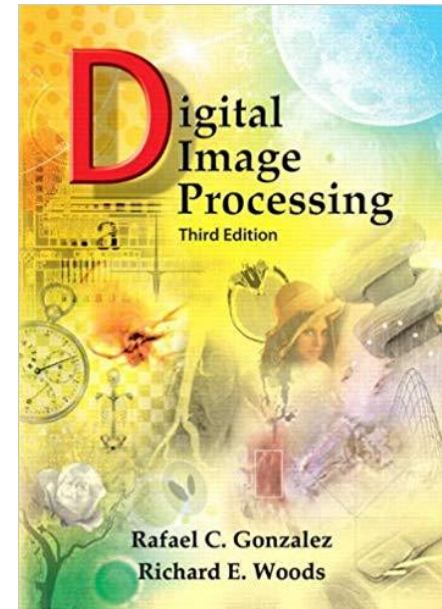
A PowerPoint presentation may be submitted instead, but it will receive a maximum of **70% credit**.

Textbook

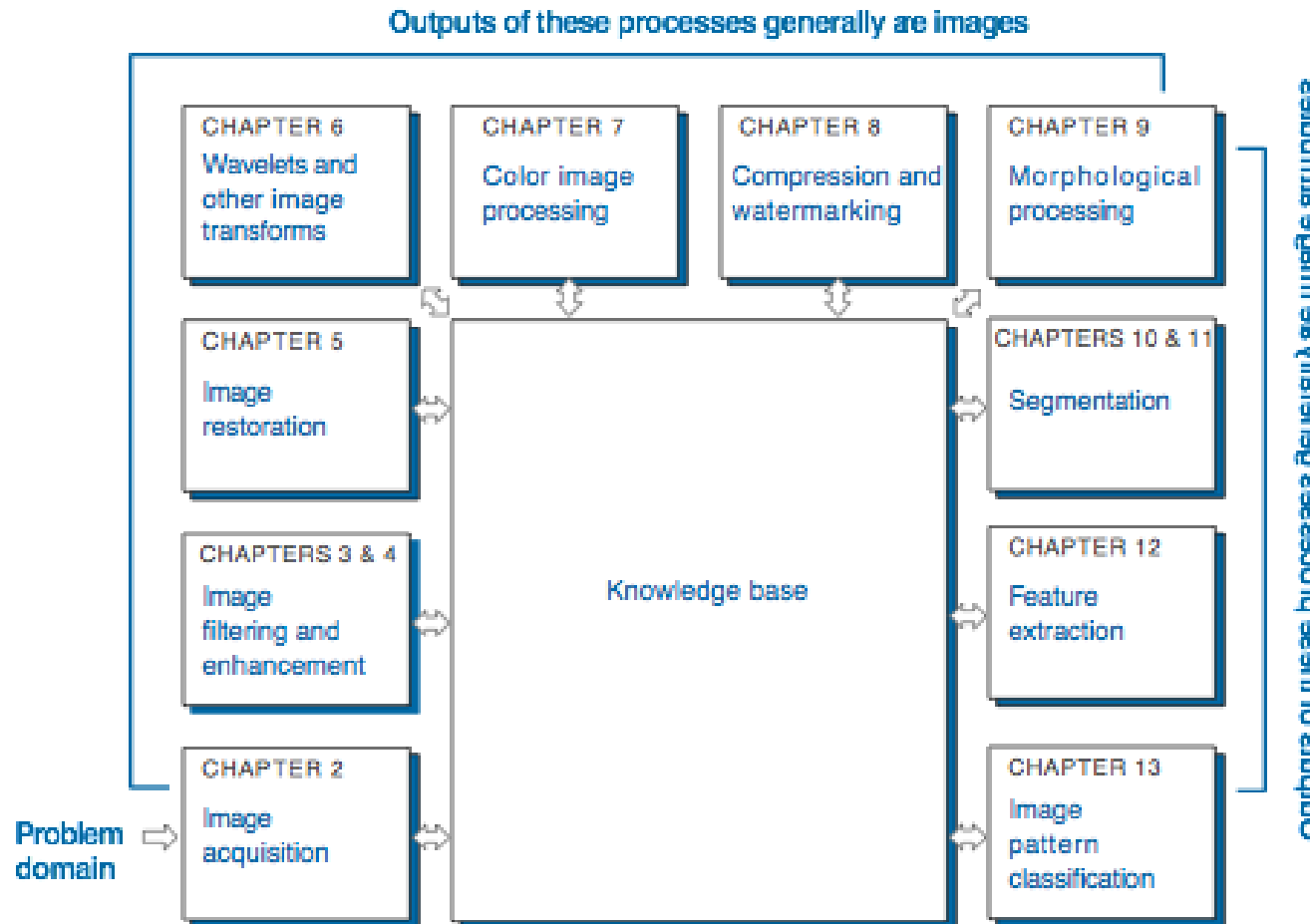
❑ Digital Image Processing by Gonzalez and Woods. 3rd or 4th edition, Pearson

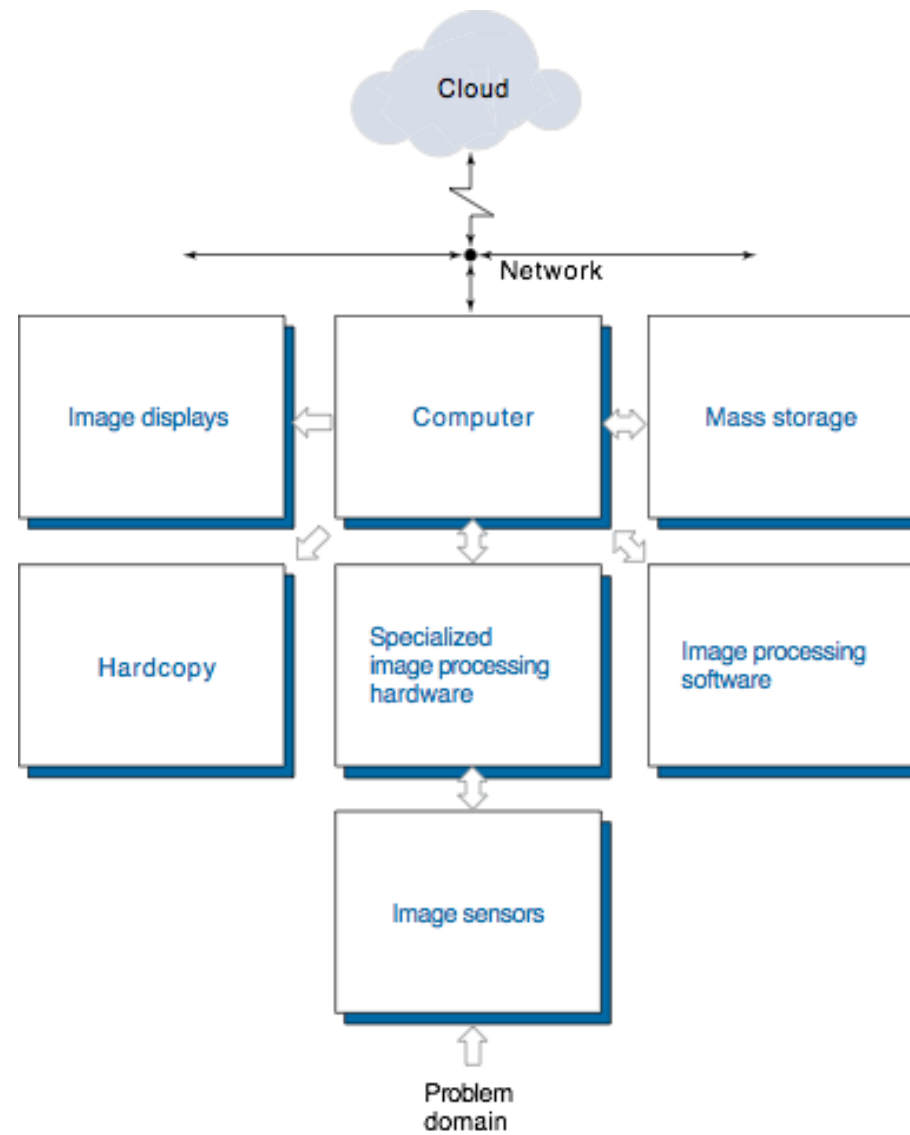
Content List:

1. Introduction
2. Digital image fundamentals
3. Intensity transformation & spatial filtering
4. Filtering in the frequency domain
5. Image restoration & reconstruction
6. Color image processing
7. Wavelets & other image transforms
8. Image compression
9. Morphological image processing
10. Image segmentation
11. Feature Extraction
12. Image Pattern Classification



Topics





Class Topics

- ☐ Introduction & syllabus
- ☐ Background and math tools in digital image processing
- ☐ Human visual system, color models
- ☐ Processing of binary/gray images
- ☐ Contrast enhancement
- ☐ Midterm Exam
- ☐ Edge detection and sharpening
- ☐ Noise removal
- ☐ Image/video compression
- ☐ Image restoration
- ☐ Final presentation

Mathematical Background Requirements

- ☐ **Linear Algebra:**

- ☐ Vector Space

- ☐ Eigenvalues & Eigenvectors

- ☐ Least-Squares Estimation

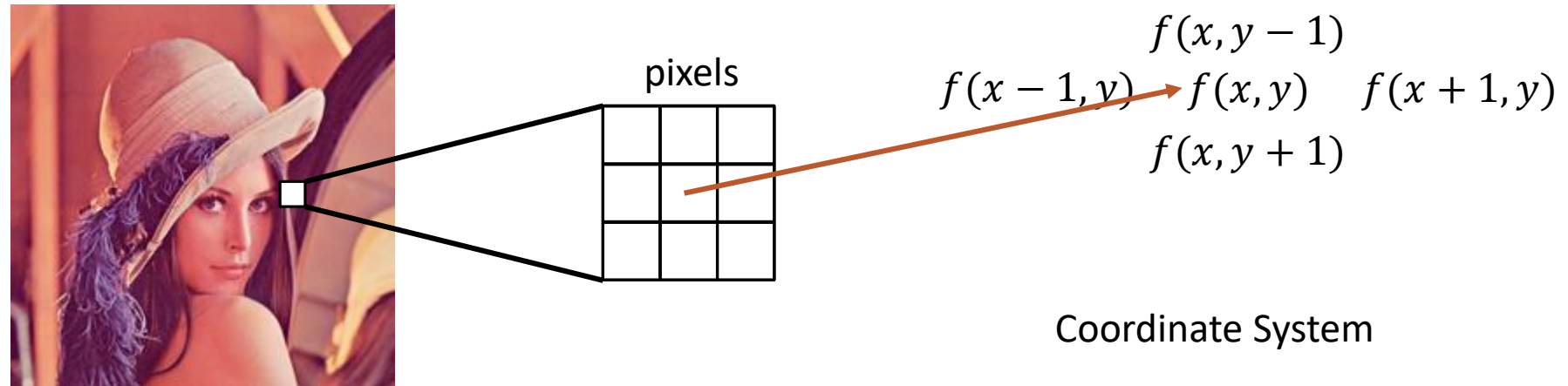
- ☐ Singular Value Decomposition (SVD)

- ☐ **Probability**

- ☐ **Fourier Transform**

What is Digital Image Processing

An **image** can be considered a function $f: R^2 \rightarrow R$, where $f(x, y)$ represents the **intensity** at position (x, y)



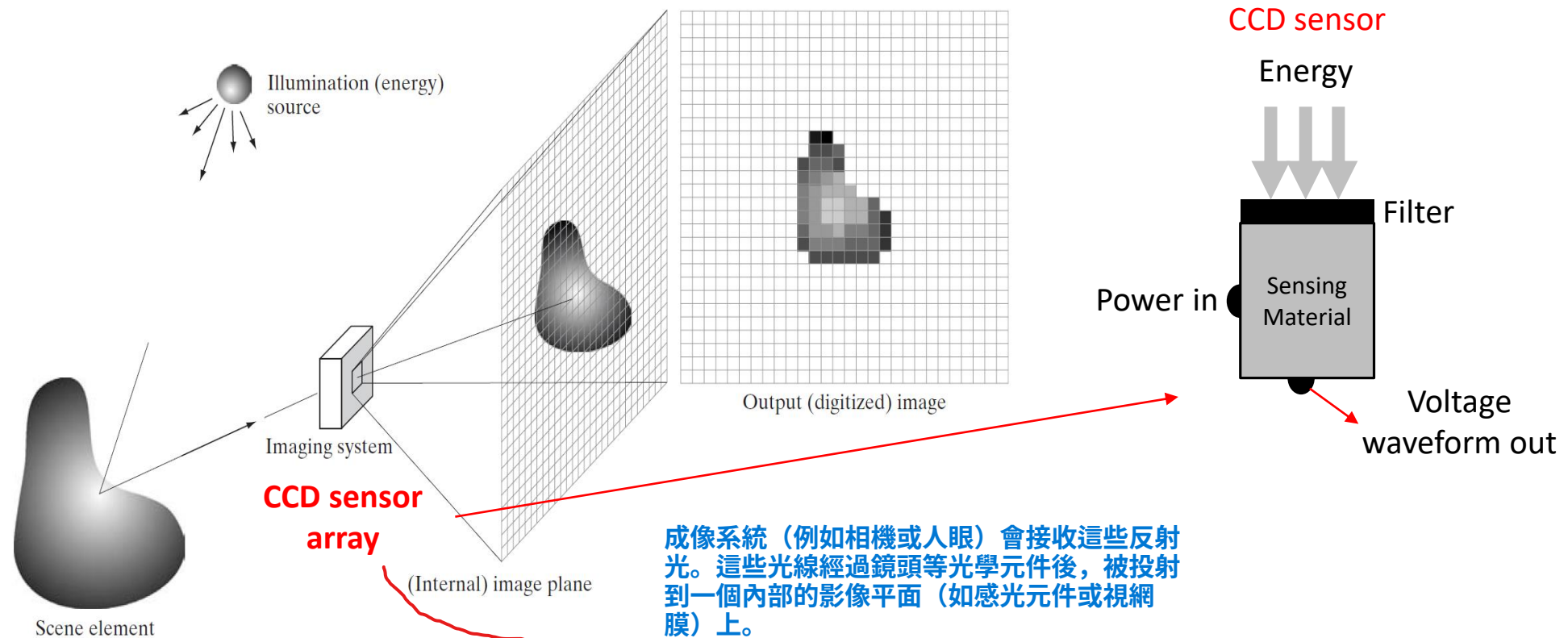
Definition of Image Processing:

Any tasks that deal with 2D digital images based on computer computation

Image Formation Process

取樣 (Sampling)：在影像平面上按照固定的間隔點取出顏色或亮度資訊，這些點就成為每個像素的中心位置。

數位化 (Digitization)：將這些取樣點的光學訊號（通常是亮度、顏色強度）轉換成數位值（例如0到255之間的整數），方便電腦處理。通常由感光元件（如CCD或CMOS感測器）完成，這些元件會將光信號轉換為電信號，再由類比數位轉換器（ADC）將其量化為數位數據。



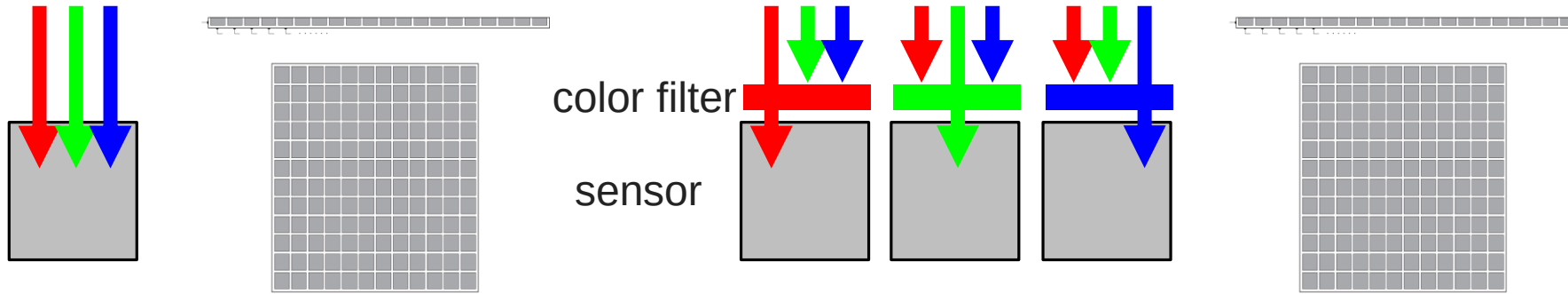
成像系統（例如相機或人眼）會接收這些反射光。這些光線經過鏡頭等光學元件後，被投射到一個內部的影像平面（如感光元件或視網膜）上。

Scene element is illuminated by a light (or energy) source, which reflects light towards the imaging system

Imaging system captures reflected light and projects it onto an internal image plane, forming a **continuous image** based on the scene's shape and lighting.

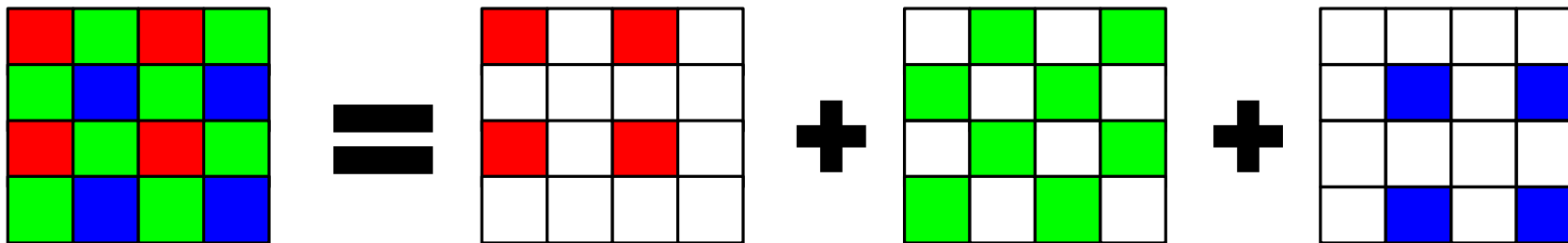
Internal image is sampled and digitized into a pixelated grid, producing a low-resolution output represented by square pixels.

CCD sensor



Monochrome sensor

Color sensor



Bayer Pattern



(a) R plane



(b) G plane



(c) B plane



A color image can be represented as three functions put together:

$$f(x, y) = f_r(x, y), f_g(x, y), f_b(x, y)$$

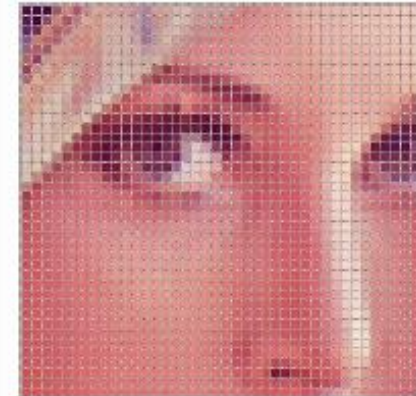
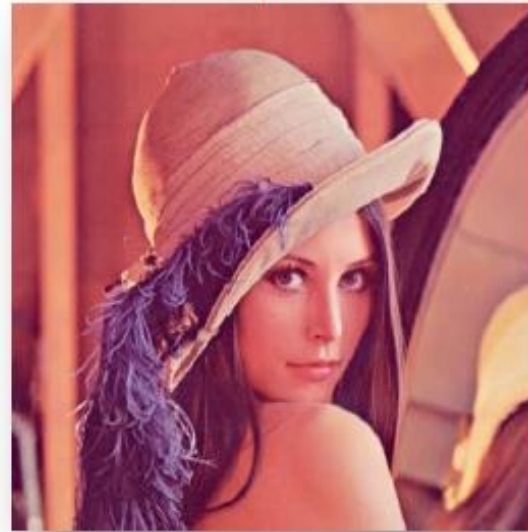
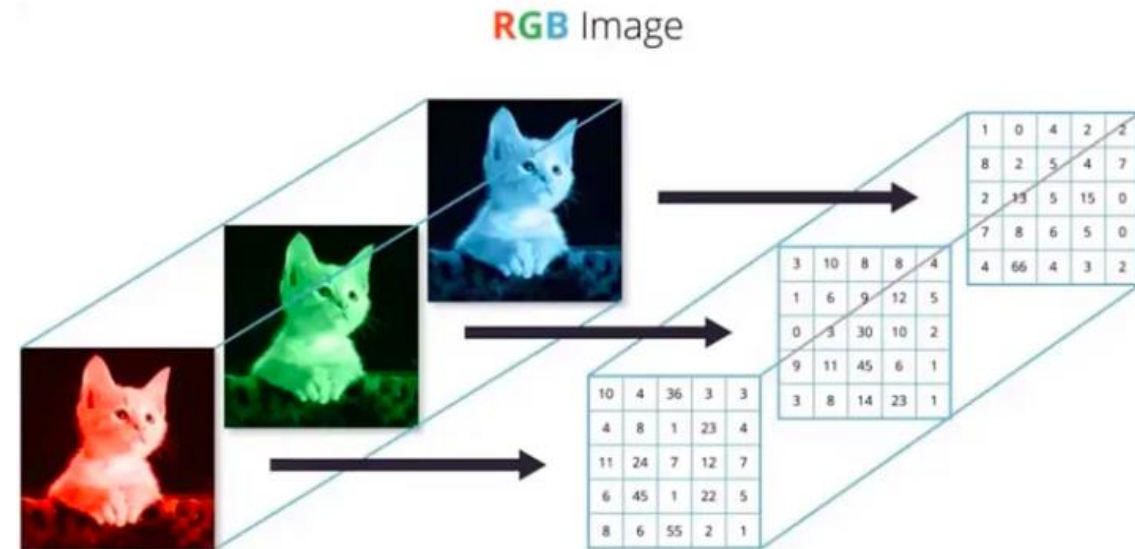
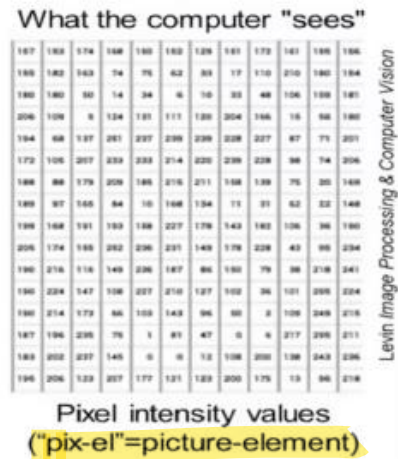
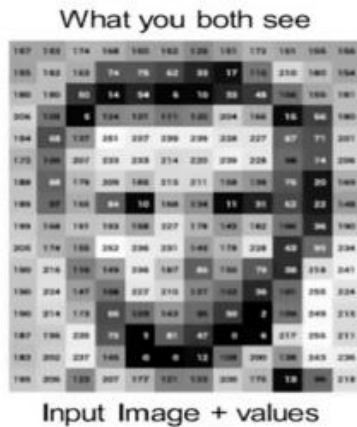
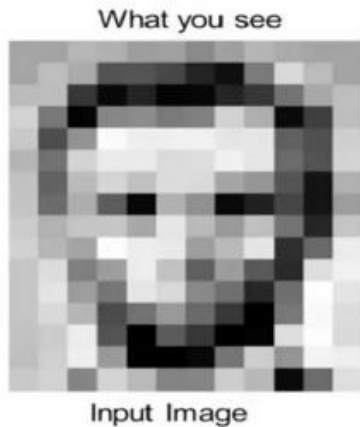


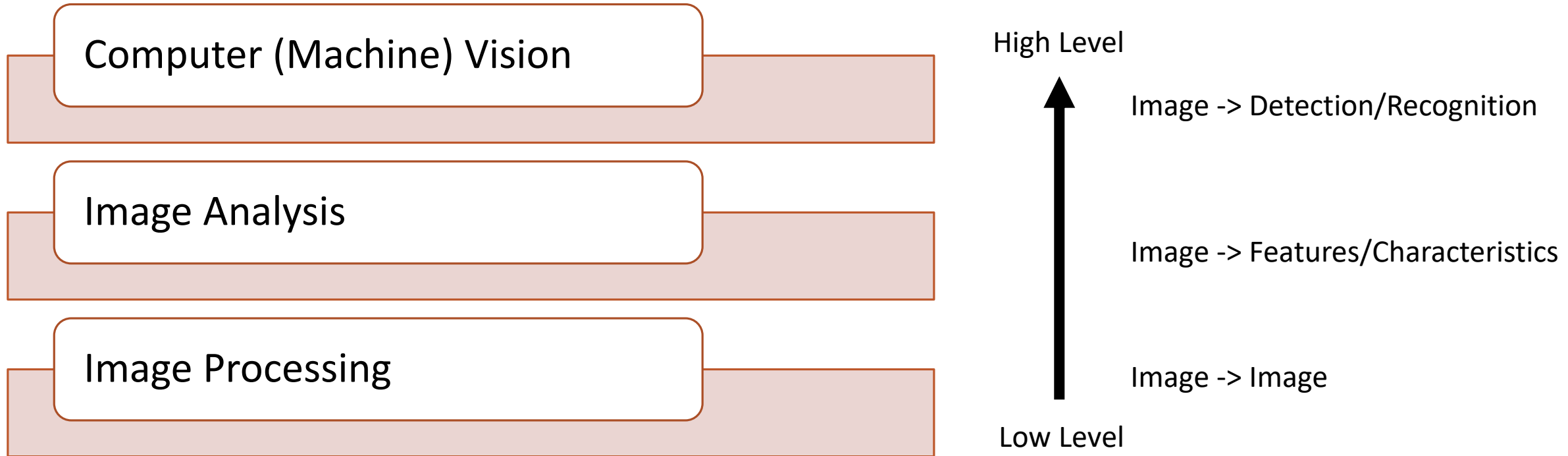
Image Representation by Computer

Images as numbers





What is Digital Image Processing



What is Computer Vision

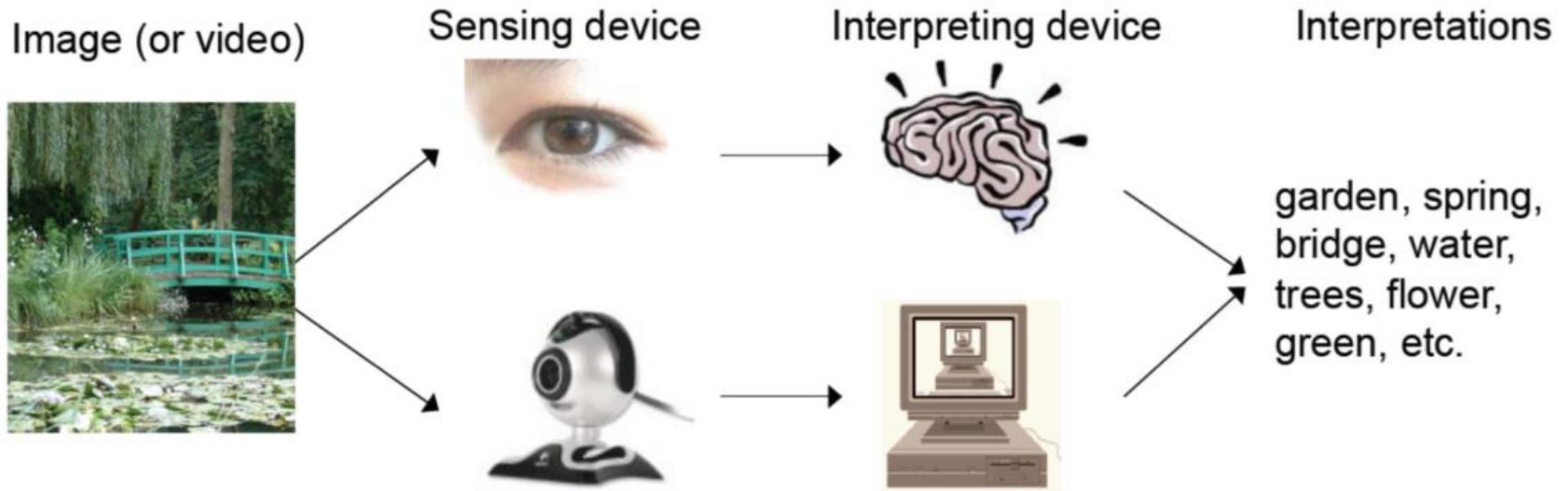


Computer Vision:

a field of artificial intelligence (AI) that enables computers and systems to derive meaningful information from digital images, videos and other visual inputs — and take actions or make recommendations based on that information.

If AI enables computers to think, computer vision enables them to see, observe and understand.

Human vision vs. Computer vision





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49	49	99	40	17	81	18	57	60	87	17	40	98	43	69	48	04	56	62	00
81	49	31	73	55	79	14	29	93	71	40	67	58	88	30	03	49	13	36	65
52	70	95	23	04	60	11	42	69	24	68	56	01	32	56	71	37	02	36	91
22	31	16	71	51	63	83	89	41	92	36	54	22	40	40	28	66	33	13	80
24	47	38	00	99	03	45	02	44	75	33	53	78	36	84	20	35	17	12	50
32	98	81	28	64	23	67	10	26	38	40	67	59	54	70	66	18	38	64	70
67	26	20	68	02	62	12	20	95	63	94	39	63	08	40	91	66	49	94	21
24	55	58	05	66	73	99	26	97	17	78	78	96	83	14	88	34	89	63	72
21	36	23	09	75	00	76	44	20	45	35	14	00	61	33	97	34	31	33	95
78	17	53	28	22	75	31	67	15	94	03	80	04	62	16	14	09	53	56	92
16	39	05	42	96	35	31	47	55	58	88	24	00	17	54	24	36	29	85	57
86	56	00	48	35	71	89	07	05	44	44	37	44	60	21	58	51	54	17	58
19	80	81	68	05	94	47	69	28	73	92	13	86	52	17	77	04	89	55	40
04	52	08	83	97	35	99	16	07	97	57	32	16	26	26	79	33	27	98	66
00	36	68	87	57	62	20	72	03	46	33	67	46	55	12	32	63	93	53	69
04	42	16	73	35	25	39	11	24	94	72	18	08	46	29	32	40	62	76	36
20	69	36	41	72	30	23	88	34	82	89	69	82	67	59	85	74	04	36	16
20	73	35	29	78	31	90	01	74	31	49	71	48	86	81	16	23	57	05	54
01	70	54	71	83	51	54	69	16	92	33	48	61	43	52	01	89	19	67	48

What the computer sees

Background of Digital Image Processing

- ❑ Image Processing basically originates from
 - ❑ Improving visual quality of images
 - ❑ Developing automatic machine vision
- ❑ History

影像處理的起源與發展：

初期目標：

提升視覺品質

讓機器可以自動理解影像（機器視覺）

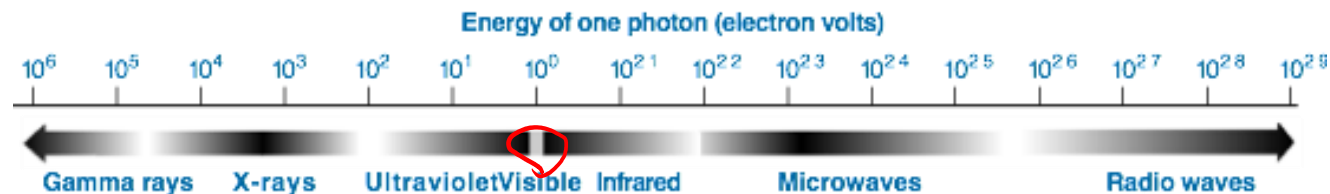
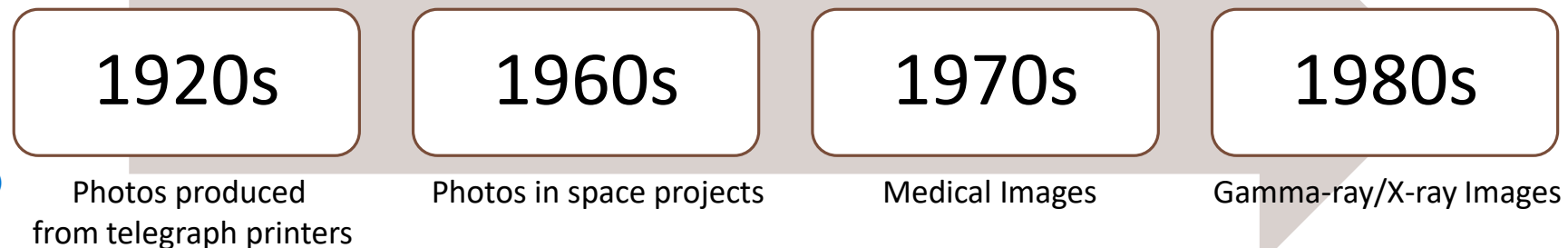
歷史發展：

1920s：電傳打字機列印照片

1960s：太空計畫中的圖像處理

1970s：醫療影像（如 CT、MRI）

1980s：工業與科研應用擴展



a b
c d

FIGURE 1.6

Examples of gamma-ray imaging.

(a) Bone scan.

(b) PET image.

(c) Cygnus Loop.

(d) Gamma radiation (bright spot) from a reactor valve.

(Images

courtesy of

(a) G.E. Medical

Systems; (b) Dr.

Michael E. Casey,

CTI PET Systems;

(c) NASA;

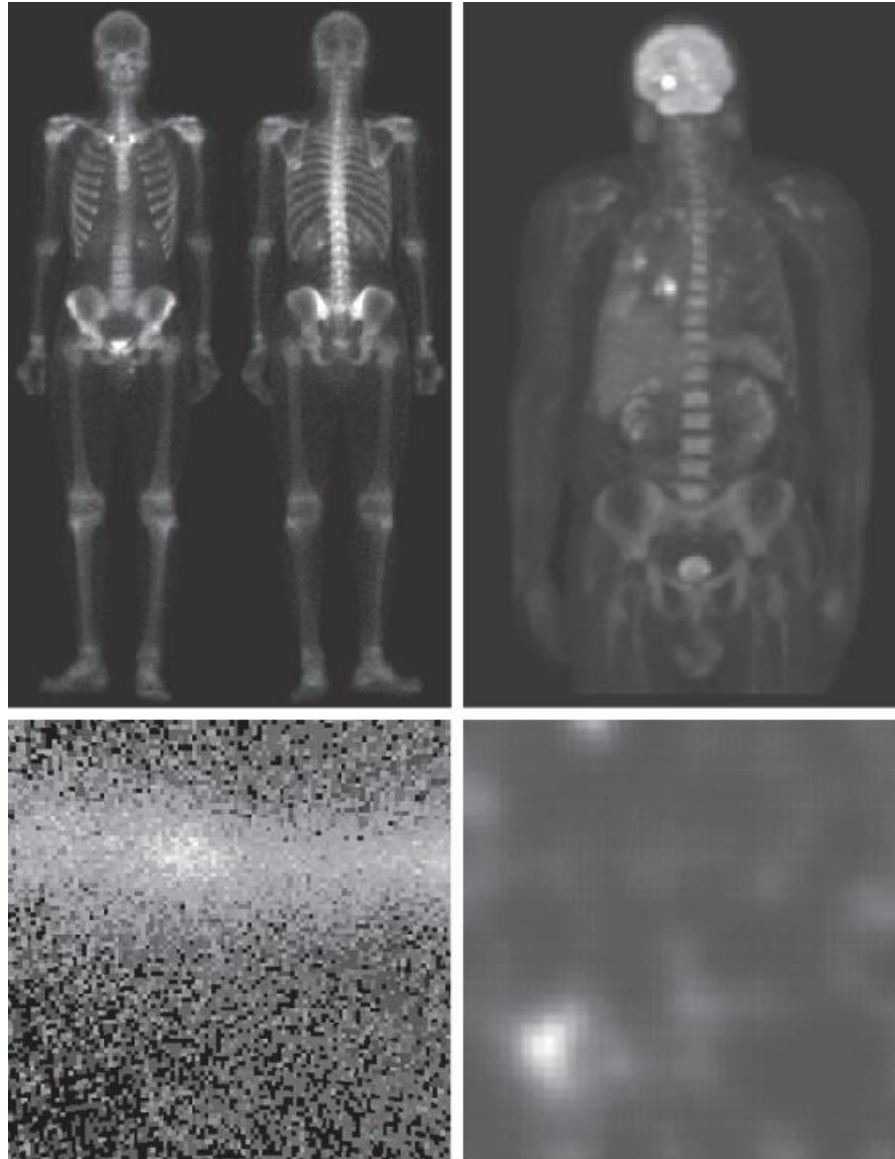
(d) Professors

Zhong He and

David K. Wehe,

University of

Michigan.)



Primary Applications:

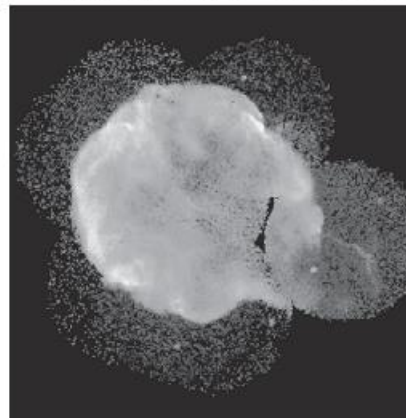
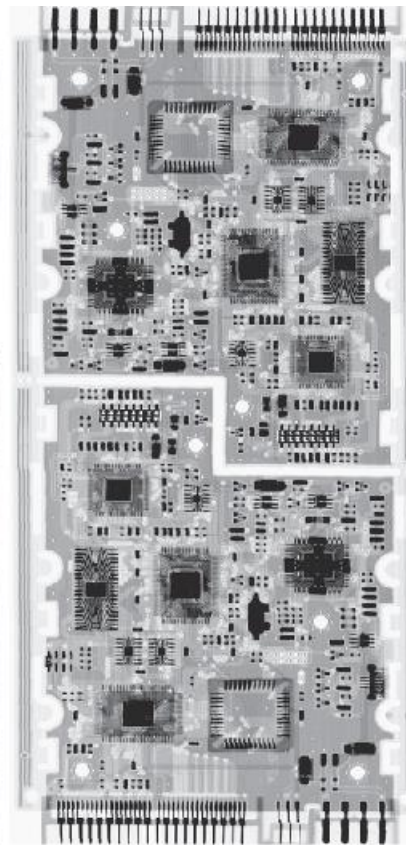
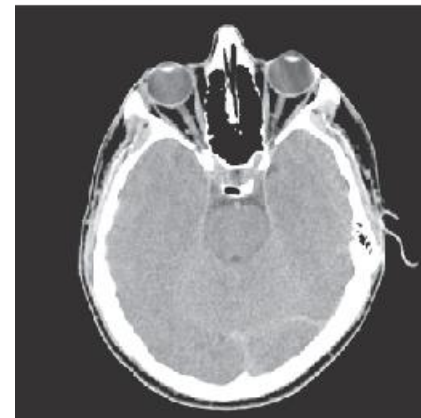
1. Medical Imaging
2. Space Imaging



FIGURE 1.1 A digital picture produced in 1921 from a coded tape by a telegraph printer with special typefaces. (McFarlane.) [References in the bibliography at the end of the book are listed in alphabetical order by authors' last names.]



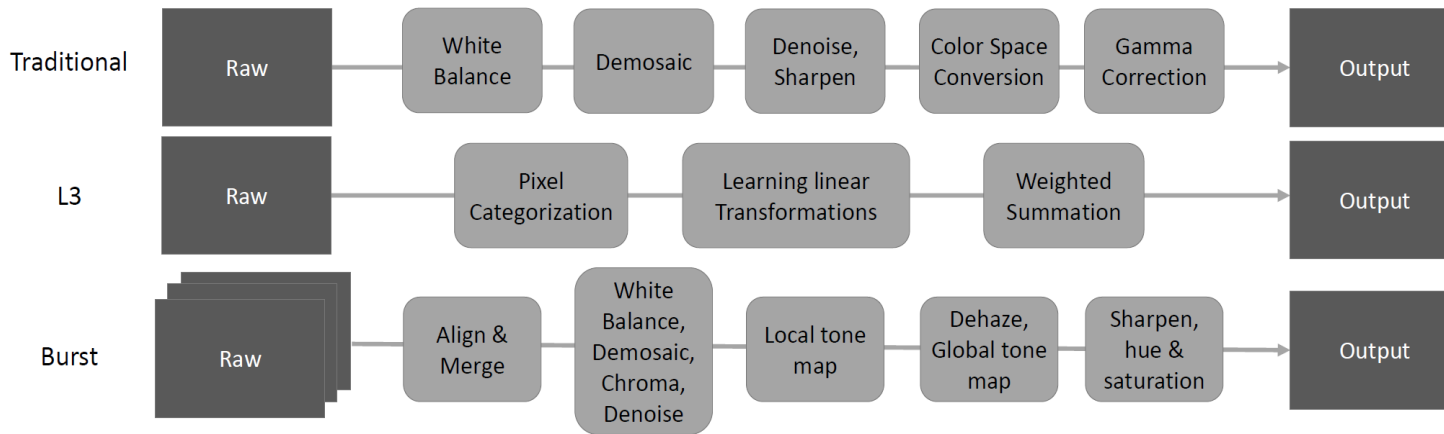
FIGURE 1.4 The first picture of the moon by a U.S. spacecraft. *Ranger 7* took this image on July 31, 1964 at 9:09 A.M. EDT, about 17 minutes before impacting the lunar surface. (Courtesy of NASA.)



a
d
c
b
e

FIGURE 1.7 Examples of X-ray imaging. (a) Chest X-ray. (b) Aortic angiogram. (c) Head CT. (d) Circuit boards. (e) Cygnus Loop. (Images courtesy of (a) and (c) Dr. David R. Pickens, Dept. of Radiology & Radiological Sciences, Vanderbilt University Medical Center; (b) Dr. Thomas R. Gest, Division of Anatomical Sciences, Univ. of Michigan Medical School; (d) Mr. Joseph E. Pascente, Lixi, Inc.; and (e) NASA.)

Image Signal Processing Pipeline



Bayer Pattern

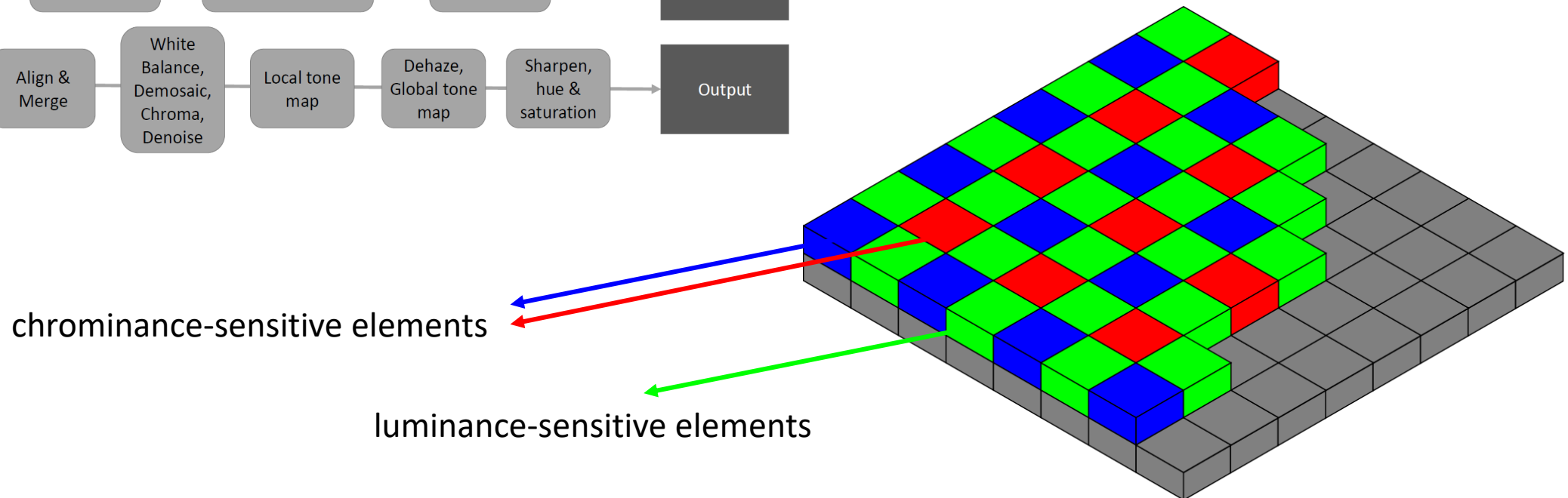
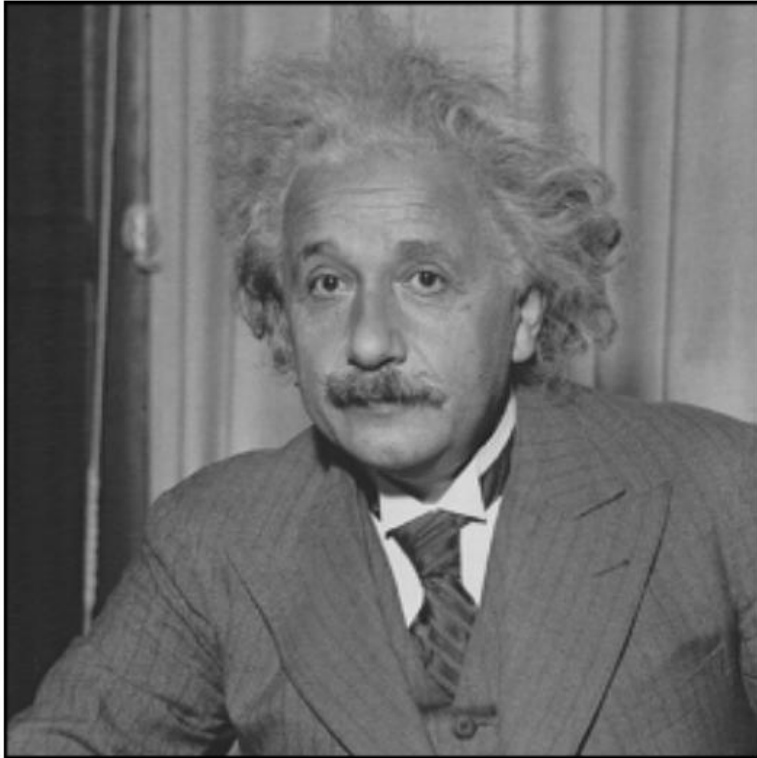
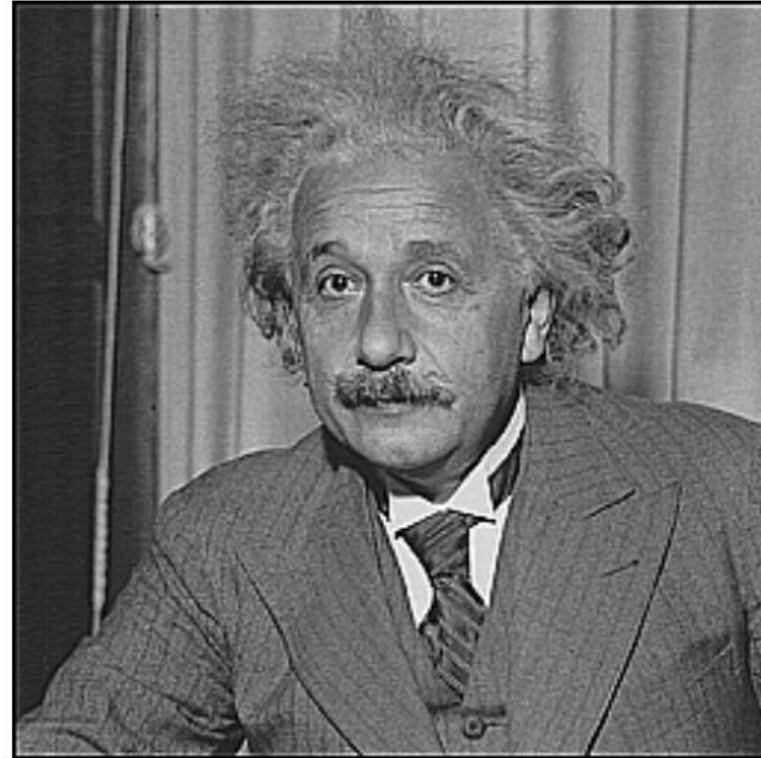


Image Sharpening



before

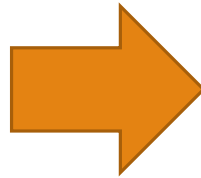


after

Contrast Enhancement



Low contrast



High contrast

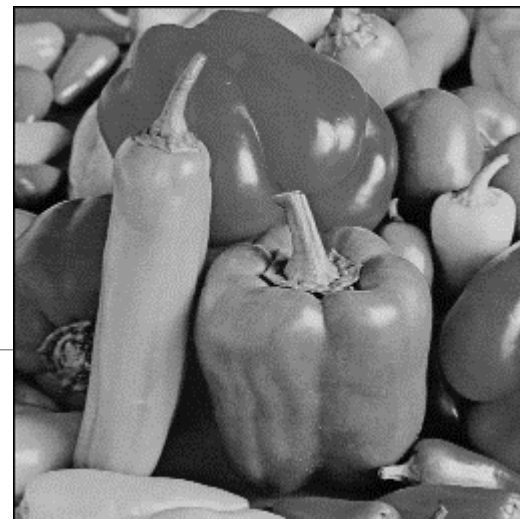
Edge Detection



Image Noise

Common Noise Types

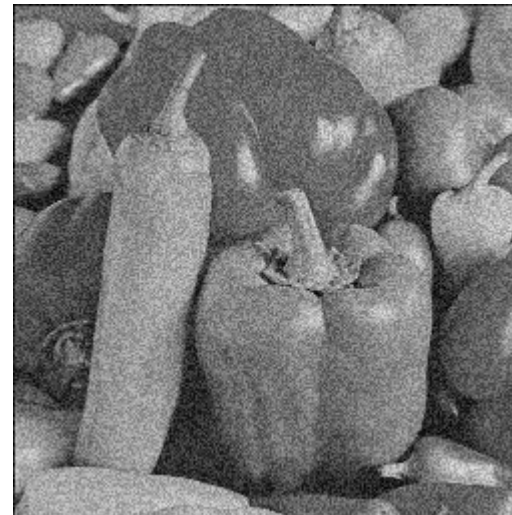
- Impulse noise
 - Salt-and-pepper noise
 - Random-valued noise
- Gaussian noise



Clean Image



Salt-and-pepper noise



Random-valued noise



Gaussian noise

Noise Removal

- How to filter/smooth out noise?

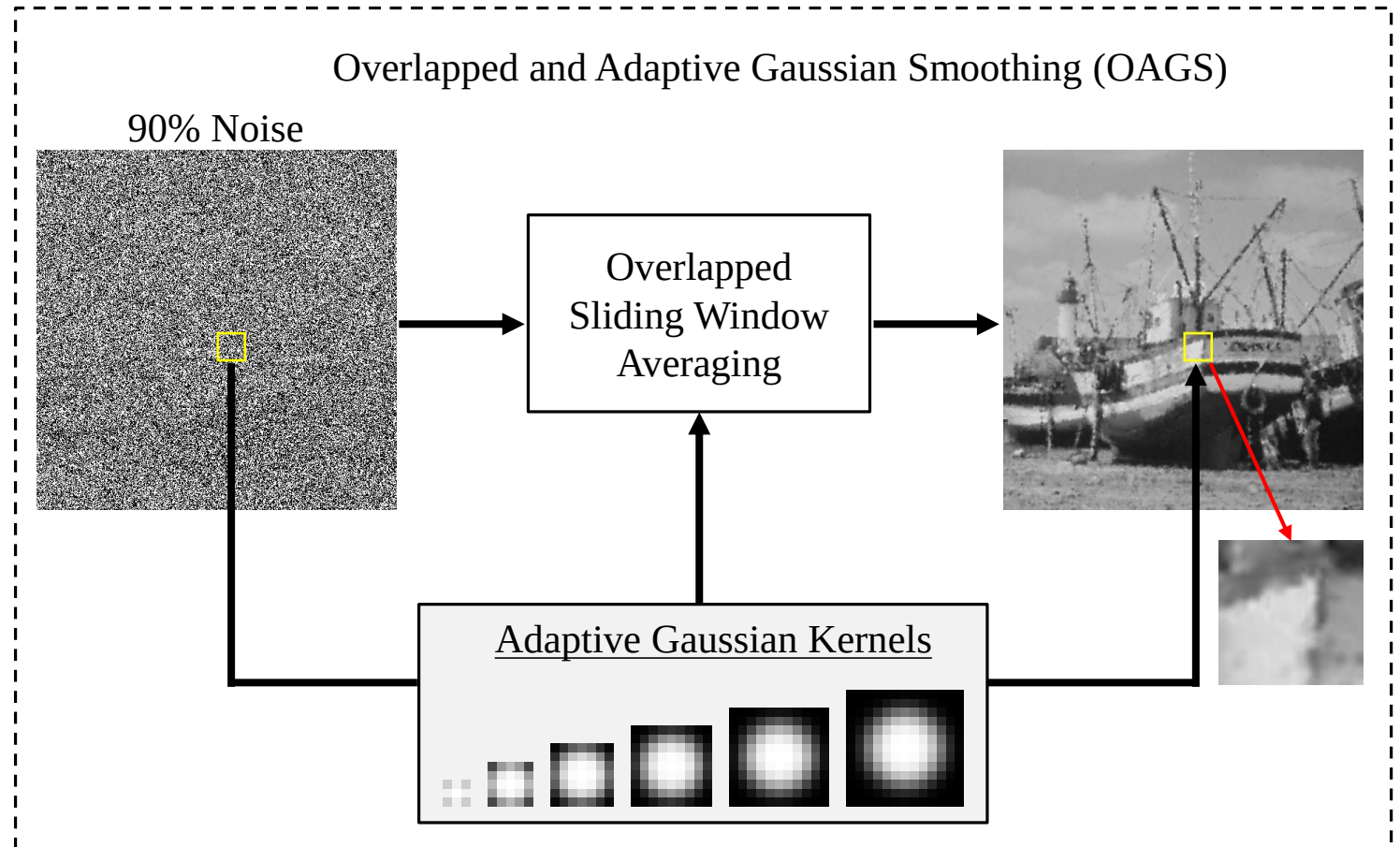


Image Compression

Example of Lossy Compression



**Original Lena Image
(12KB size)**

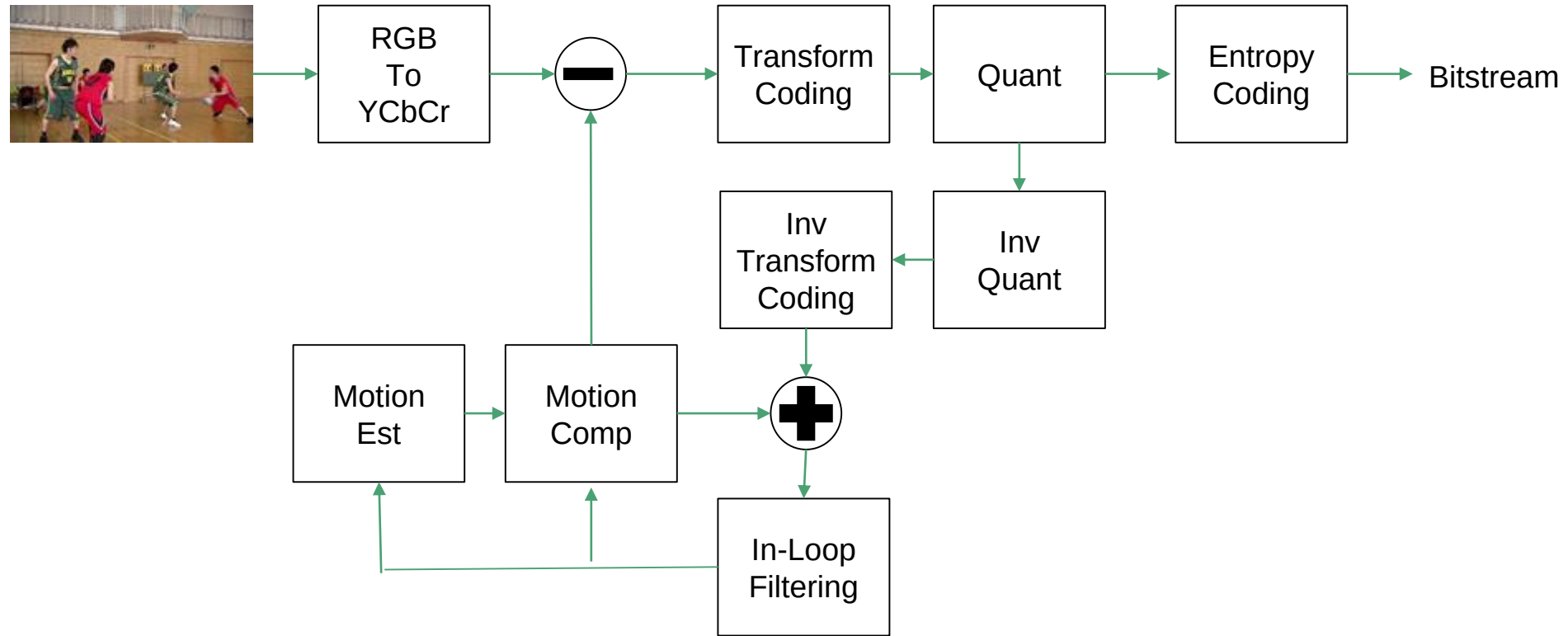


**Lena Image,
Compressed (85%
less information,
1.8KB)**



**Lena Image, Highly
Compressed (96%
less information,
0.56KB)**

Video Compression



60 fps 4K uncompressed video = $4096 * 2160 * 24 \text{ bit} * 60 \text{ frame/sec} \approx 11.87 \text{ Gbps}$

影像復原：受損圖像

Image Restoration - Degraded Images



Image Restoration - Restored Images



Underwater Image Restoration



不同曝光時間所拍攝
的多張影像可合成一
張 HDR 圖像

可同時保留：

暗部細節（短曝光）

亮部細節（長曝光）

High Dynamic Range (HDR) Imaging

short Exposure time long



short Exposure time long

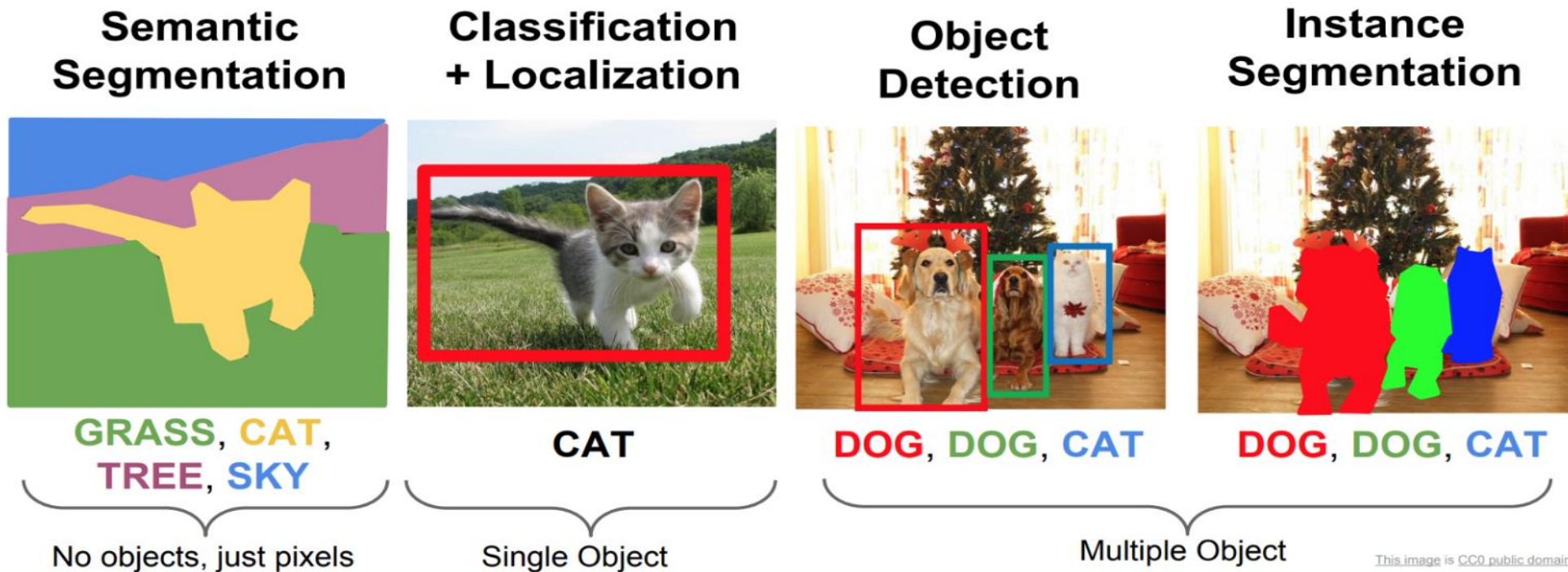


Method A

Method B

Method C

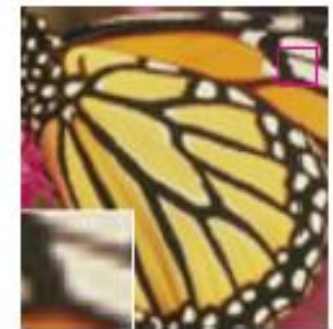
Object Detection & Recognition



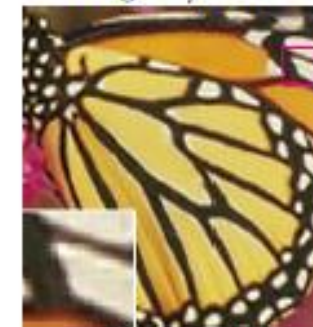
Super Resolution



Original / PSNR



Bicubic / 24.04 dB



SC / 25.58 dB



SRCNN / 27.95 dB

Image Deblurring

Image deblurring is a low-level vision task to remove blurring effects from images



Image Deraining

Image deraining is a low-level vision task to rain streaks from images



摩爾紋 (Moiré pattern) : 拍攝電子螢幕或條紋時常見的干擾圖樣

Image Demoiréing

Moiré patterns can be possibly found in images taken of a digital display, where the two arrangements are superimposed in forming the color and shape irregularities, causing repetitive delicate patterns.



Input

Demoiréed

Python with OpenCV

- ❑ Python

 - ❑ <https://www.python.org/>

- ❑ Anaconda

 - ❑ <https://www.anaconda.com/>

- ❑ OpenCV

 - ❑ <https://pypi.org/project/opencv-python/>

Basic Image Operations

- ❑ Read an image
 - ❑ `img = cv2.imread(filename, [flags])`
 - ❑ commonly-used flags
 - ❑ `cv2.IMREAD_GRAYSCALE`
 - ❑ `cv2.IMREAD_COLOR`
- ❑ Show an image
 - ❑ `cv2.imshow(window_name, img)`
- ❑ Destroy a specific window that shows an image (to release the resource)
 - ❑ `cv2.destroyWindow(window_name)`
- ❑ Save an image to a file
 - ❑ `retval = cv2.imwrite(filename, img, [params])`